

GUJARAT UNIVERSITY - MID TERM EXAM 2026

Subject: Computer Science & Operating Systems | Max Marks: 20

OFFICIAL MODEL ANSWER KEY & MARKING RUBRIC

Q1 Rubric (RAM vs ROM - 3 Marks):

- RAM is Volatile (loses data on power loss); ROM is Non-Volatile (permanent). (1.5 Marks)
- RAM supports read/write; ROM is typically read-only (BIOS/Firmware). (1.5 Marks)

Q2 Rubric (CPU Components - 3 Marks):

- ALU (Arithmetic Logic Unit): Arithmetic and logical calculations. (1 Mark)
- Control Unit (CU): Direction, synchronization, decoding of instructions. (1 Mark)
- Registers / Cache: Ultra-fast temporary internal registers. (1 Mark)

Q3 Rubric (OS Functions - 7 Marks):

- Process Management & CPU scheduling. (2 Marks)
- Memory Management (RAM allocation, paging, virtual memory). (2 Marks)
- File System Management & Directory indexing. (1.5 Marks)
- Device / I/O handling & Security protection. (1.5 Marks)

Q4 Rubric (Process vs Thread - 7 Marks):

- Process: Independent program in execution with separate memory space. Heavyweight. (3.5 Marks)
- Thread: Lightweight sub-process sharing the same process address space and resources. (3.5 Marks)